



Teacher Availability Configuration Guide

Overview

The **Teacher Availability** configuration page has been enhanced with **3 distinct options** to manage teacher constraints and preferences during lab scheduling optimization.



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Option 1: Days per week unavailable



Description

Block **entire weekdays** when a teacher is never available.



Type

Hard constraint - The teacher will NEVER be assigned on these days.

Use cases

- Teacher doesn't work on specific weekdays (e.g., "Never works on Mondays")
- Part-time teachers with fixed unavailable days
- Teachers with recurring commitments on specific weekdays

How to configure

1. Select the teacher from the dropdown
2. Select one or more unavailable weekdays (Lunes, Martes, Miércoles, Jueves, Viernes)
3. Click "Add"

Example

Teacher: Roberto Fernández
Unavailable weekdays: Lunes, Viernes
→ Roberto will NEVER be scheduled on Mondays or Fridays

Data structure

```
st.session_state.advanced_config['teacher_unavailable_weekdays'] = {
    'Roberto Fernández': ['Lunes', 'Viernes'],
    'Beatriz Portillo': ['Miércoles']
}
```

Option 2: Specific day/time slots unavailable



Description

Block **specific time slots** on specific days when a teacher cannot attend.



Type

Hard constraint - The teacher will NEVER be assigned to these exact slots.

Use cases

- Teacher has a recurring meeting every Tuesday 10:30-12:30
- Teacher teaches another course at specific times
- Temporary unavailability on specific days

How to configure

1. Select the teacher from the dropdown
2. Select the day (Lunes, Martes, Miércoles, Jueves, Viernes)
3. Select the time slot or “(All day)” to block the entire day
4. Click “Add”

Available time slots

- 08:30-10:30
- 10:30-12:30
- 12:30-14:30
- 15:00-17:00
- 17:00-19:00
- 19:00-21:00
- **(All day)** - blocks all 6 time slots

Example

Teacher: Daniel Alvarez
 Unavailable: Martes 10:30-12:30, Jueves 15:00-17:00
A Daniel will NEVER be scheduled on:
 - Tuesday 10:30-12:30
 - Thursday 15:00-17:00

Data structure

```
st.session_state.advanced_config['teacher_unavailability'] = {
    'Daniel Álvarez': [
        'Martes 10:30-12:30',
        'Jueves 15:00-17:00'
    ]
}
```

Option 3: Preferred time range

★ Description

Define **soft preferences** for when a teacher prefers to work.

✓ Type

Soft constraint - The optimizer will TRY to respect these but won't block if impossible.

Use cases

- Teacher prefers morning slots (08:30-12:30)
- Teacher prefers to work maximum 3 days per week
- Teacher prefers afternoon slots (15:00-19:00)

How to configure

1. Select the teacher from the dropdown
2. Set "Max days/week" (0 = no limit, 1-5 = preferred maximum)
3. Select preferred time slots (multiple selection allowed)
4. Click "Set"

Features

- **Max days/week:** Soft limit on the number of lab days per week
 - 0 = no limit
 - 1-5 = preferred maximum (warns if exceeded but doesn't block)
- **Preferred time slots:** Time slots the teacher prefers
 - Optimizer penalizes assignments outside this range
 - Never blocks if impossible to satisfy

Example

```
Teacher: Alessio Zuliani
Max days/week: 3
Preferred time slots: 08:30-10:30, 10:30-12:30
☐ Optimizer will:
  - Try to assign ≤ 3 days/week
  - Prefer morning slots (08:30-12:30)
  - Still assign if necessary even if it exceeds preferences
```

Data structure

```
st.session_state.advanced_config['teacher_preferences'] = {
    'Alessio Zuliani': {
        'max_days_per_week': 3,
        'preferred_blocks': [1, 2] # Block indices (1-indexed)
    }
}
```

Comparison: Hard vs Soft Constraints

Feature	Hard Constraint 	Soft Constraint 
Options	Option 1, Option 2	Option 3
Behavior	NEVER violated	Optimizer tries to respect
If impossible	Schedule fails or finds alternative	Schedule continues with penalty
Use case	Absolute unavailability	Preferences
Impact	Blocks assignments	Penalizes but allows

Priority order

1. **Hard constraints** (Options 1 & 2) are checked first → Must be satisfied
2. **Soft constraints** (Option 3) are applied → Optimizer tries to minimize violations

Usage Examples

Example 1: Part-time teacher

Teacher: Maria García
 Option 1: Unavailable weekdays = Lunes, Martes, Miércoles
 Option 3: Max days/week = 2
 → Maria only works Thu/Fri, maximum 2 days per week

Example 2: Teacher with recurring meeting

Teacher: Juan Pérez
 Option 2: Martes 10:30-12:30 (hard block)
 Option 3: Preferred slots = 08:30-10:30, 15:00-17:00
 → Juan is blocked on Tuesday 10:30-12:30, prefers early morning/afternoon

Example 3: Morning-only teacher

Teacher: Ana López
 Option 3: Preferred time slots = 08:30-10:30, 10:30-12:30, 12:30-14:30
 Option 3: Max days/week = 4
 → Ana prefers mornings, maximum 4 days/week (soft)

Example 4: Complex availability

Teacher: Carlos Ruiz
 Option 1: Viernes (never available)
 Option 2: Lunes 19:00-21:00, Miércoles 17:00-19:00 (hard blocks)
 Option 3: Max days/week = 3, Preferred = 10:30-12:30, 12:30-14:30
 → Carlos:
 - Never on Friday
 - Blocked Monday evening, Wednesday afternoon
 - Prefers 3 days/week, midday slots

Technical Implementation

Session state structure

```
st.session_state.advanced_config = {
    # Option 1: Days per week unavailable (hard)
    'teacher_unavailable_weekdays': {
        'Teacher Name': ['Lunes', 'Viernes', ...]
    },

    # Option 2: Specific day/time unavailable (hard)
    'teacher_unavailability': {
        'Teacher Name': [
            'Martes 10:30-12:30',
            'Jueves 15:00-17:00',
            ...
        ]
    },

    # Option 3: Preferences (soft)
    'teacher_preferences': {
        'Teacher Name': {
            'max_days_per_week': 3,          # int or None
            'preferred_blocks': [1, 2, 3]    # block indices (1-6)
        }
    }
}
```

Block indices mapping

```
Block index 📅 Time slot
1 📅 08:30-10:30
2 📅 10:30-12:30
3 📅 12:30-14:30
4 📅 15:00-17:00
5 📅 17:00-19:00
6 📅 19:00-21:00
```

Integration with optimizer

The configuration is saved in `advanced_config` and must be integrated with `pipeline.py` during optimization:

```
def apply_teacher_constraints(model, ...):
    config = st.session_state.advanced_config

    # Apply hard constraints (Option 1)
    for teacher, unavailable_days in config.get('teacher_unavailable_weekdays', {}).items():
        # Block all slots on these days for this teacher
        pass

    # Apply hard constraints (Option 2)
    for teacher, unavailable_slots in config.get('teacher_unavailability', {}).items():
        # Block specific day/time slots for this teacher
        pass

    # Apply soft constraints (Option 3)
    for teacher, prefs in config.get('teacher_preferences', {}).items():
        if prefs.get('max_days_per_week'):
            # Add penalty if exceeds max days
            pass
        if prefs.get('preferred_blocks'):
            # Add penalty for assignments outside preferred blocks
            pass
```

Benefits



Clear separation

- **Option 1:** Weekday-level blocking
- **Option 2:** Slot-level blocking
- **Option 3:** Preferences



Flexibility

- Combine hard and soft constraints per teacher
- Multiple teachers can have different configurations



User-friendly

- Visual icons and emojis for clarity
- Expandable sections to reduce clutter

- Clear descriptions of hard vs soft

✓ Robust

- Default values ensure no errors if config missing
- Validation before adding constraints
- Easy removal of configured constraints

Backward Compatibility

Old configuration (deprecated)

The old `teacher_rules` has been replaced by:

- `teacher_unavailable_weekdays` (new - Option 1)
- `teacher_unavailability` (existing - Option 2)
- `teacher_preferences` (replaces `teacher_rules` - Option 3)

Migration

If you have old configurations, they should be migrated:

```
# Old structure
st.session_state.advanced_config['teacher_rules'] = {
    'Teacher': {
        'max_days_per_week': 3,
        'preferred_blocks': [1, 2]
    }
}

# New structure (migrated to Option 3)
st.session_state.advanced_config['teacher_preferences'] = {
    'Teacher': {
        'max_days_per_week': 3,
        'preferred_blocks': [1, 2]
    }
}
```

Future Enhancements

Possible additions

1. **Import/Export** teacher configurations as CSV
 2. **Bulk operations** for multiple teachers
 3. **Templates** for common availability patterns
 4. **Calendar view** to visualize teacher availability
 5. **Conflict detection** before running optimization
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Support

For questions or issues:

1. Check the in-app tooltips (hover over  icons)
 2. Consult this guide
 3. Contact the system administrator
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